



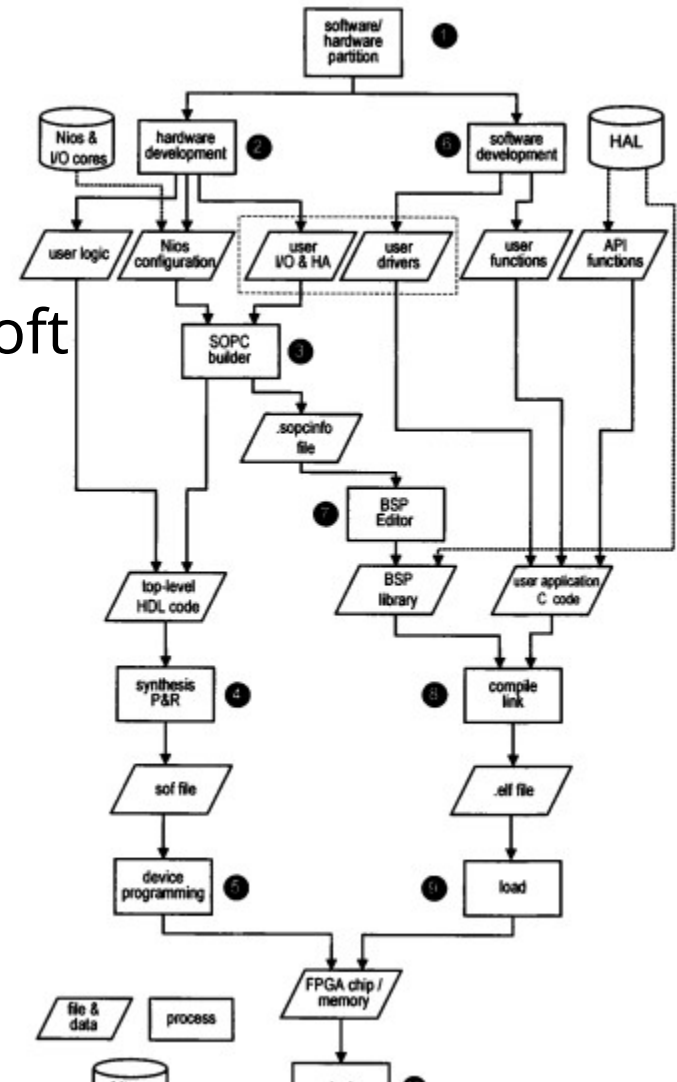
# Sexto Laboratorio

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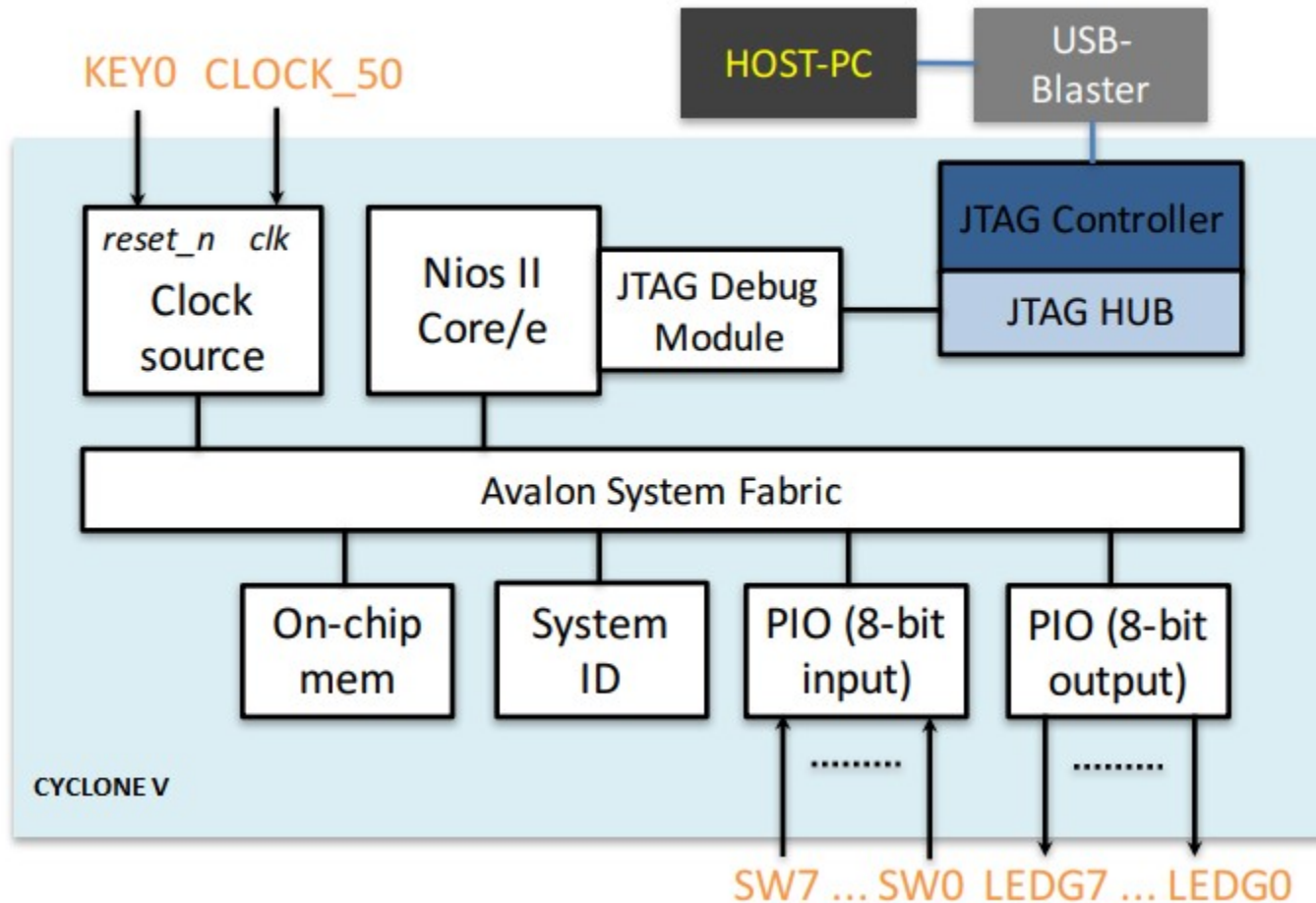
- Integración de módulos de hardware programable
- Uso de verilog
- Systembuilder/Quartus
  
- Desarrollo de un sistema con módulos de hardware programable
  
- **Construcción un SoPC**

# Construcción de un SoPC

- Particionar tareas
- Desarrollar el hardware e integrarlo con el procesador soft (NIOSEII)
- Desarrollar el software
- Implementar el software y el hardware y realizar el testing



# Primer SoPC





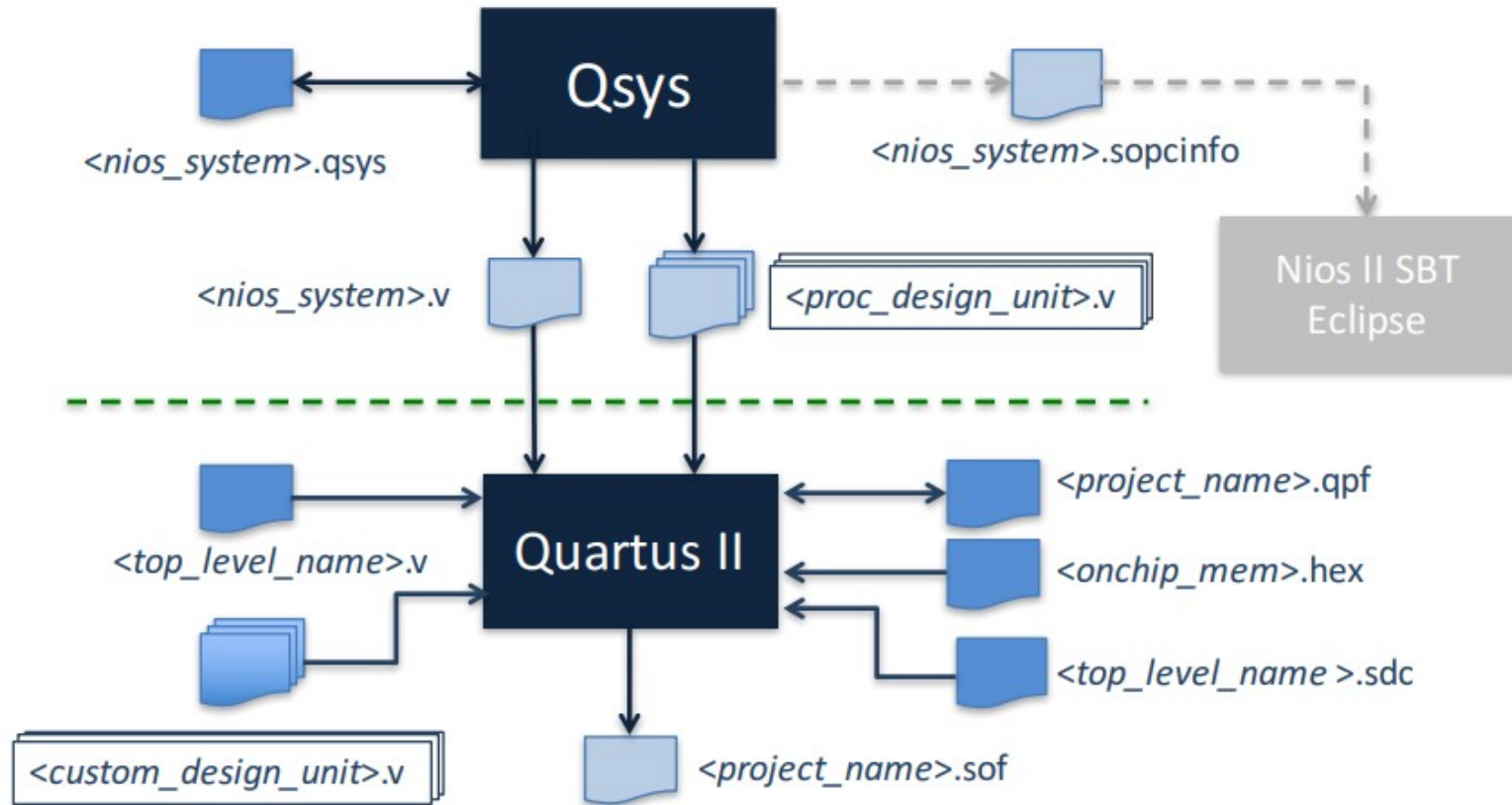
## Componentes del SoPC

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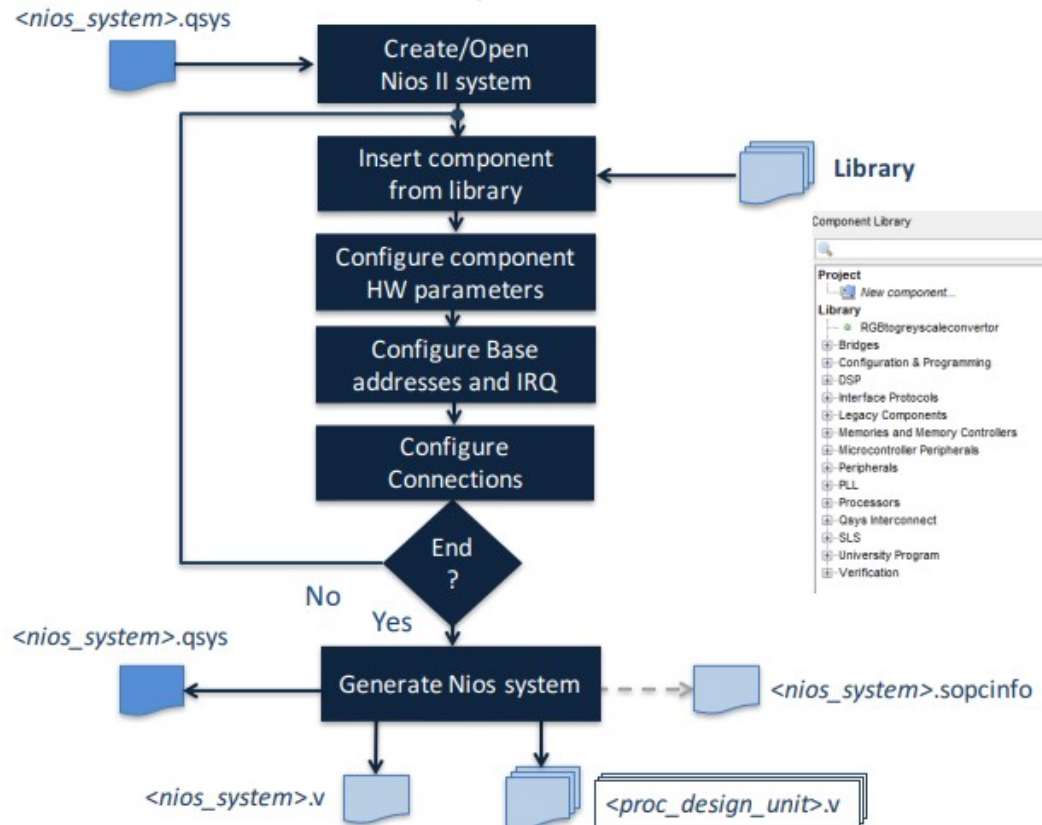
- Cpu NiosII (la version más económica) más JTAG Debug
- Onchip memoria para programa y datos (20K)
- 2 Puertos I/O
  - Input para leer los switches
  - Output para leer los leds
- System Id para identificación



# Platform Designer (Ex-Qsys)

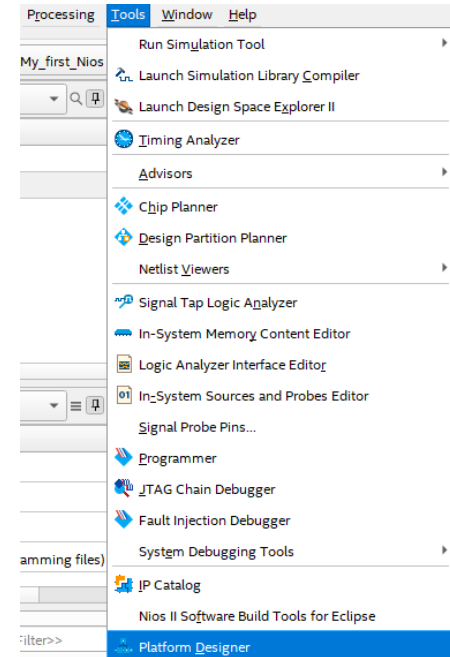


# Platform Designer (Ex-Qsys)



# Creación System Hardware

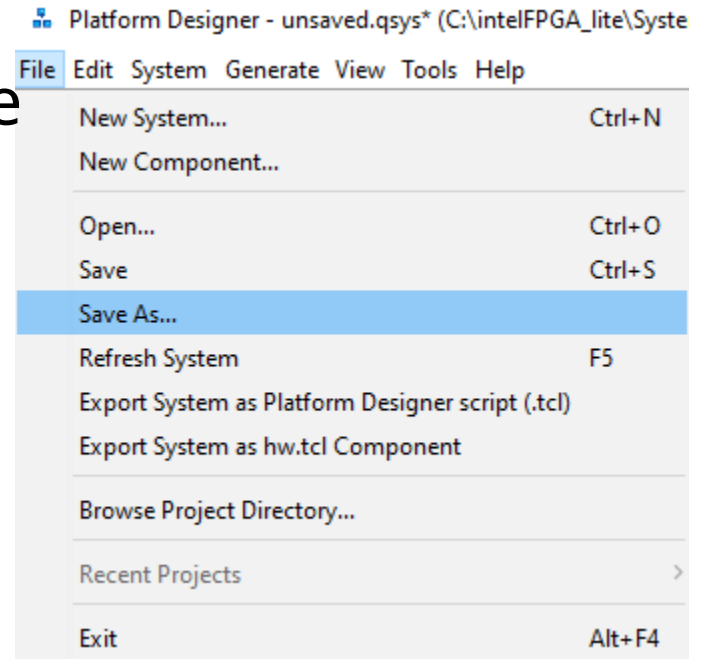
- Abrir Quartus después de crear el template con SystemBuilder
- Ir a Tools-> Platform Designer



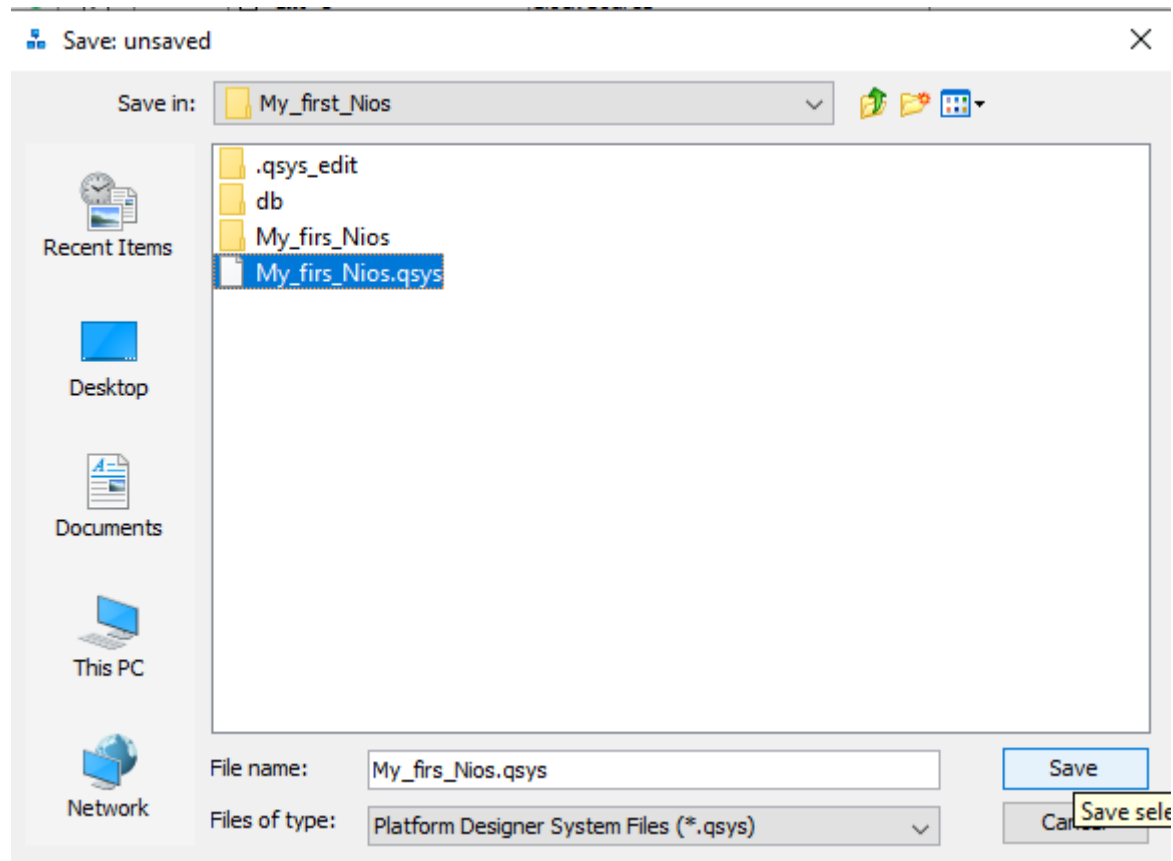


# Creación System Hardware

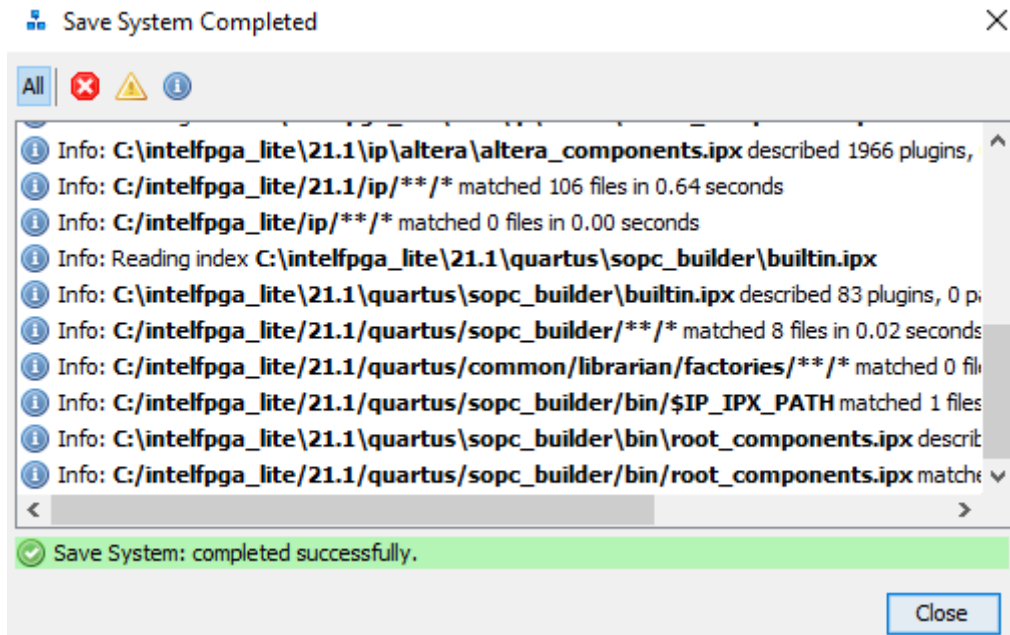
- Guardar el archivo .qsys que vamos a crear



# Creación System Hardware



# Creación System Hardware



# Creación System Hardware

## Configuración de conexiones internas

| Use                                 | Conn... | Description  | Name         | Export                 | Clock    | Base |
|-------------------------------------|---------|--------------|--------------|------------------------|----------|------|
| <input checked="" type="checkbox"/> |         | Clock Source | clk_0        |                        |          |      |
|                                     |         | Clock Input  | clk_in       | clk                    | exported |      |
|                                     |         | Reset Input  | clk_in_reset | reset                  |          |      |
|                                     |         | Clock Output | clk          | Double-click to export | clk_0    |      |
|                                     |         | Reset Output | clk_reset    | Double-click to export |          |      |

Variables que serán exportadas

Direcciones base de sistema

# Creación System Hardware

System Contents | Address Map | Interconnect Requirements

System: My\_firs\_Nios Path: clk\_0

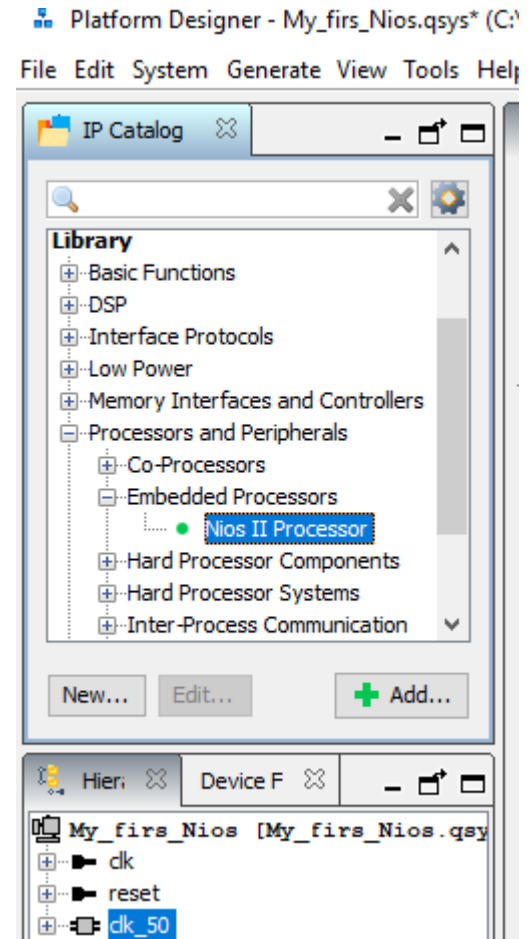
| Use                                 | Conn... | Name         | Description  | Export              |
|-------------------------------------|---------|--------------|--------------|---------------------|
| <input checked="" type="checkbox"/> |         | clk_0        | Clock Source |                     |
|                                     |         | clk_in       | Clock Input  | clk                 |
|                                     |         | clk_in_reset | Reset Input  | reset               |
|                                     |         | clk          | Clock Output | <i>Double-click</i> |
|                                     |         | clk_reset    | Reset Output | <i>Double-click</i> |

**clk\_0.clk\_reset**  
Reset Output [reset\_source 21.1]  
Associated clock: None (asynchronous)  
Synchronous edges: None



# Creación System Hardware

- Sumar un Nios II Processor, versión económica



# Creación System Hardware

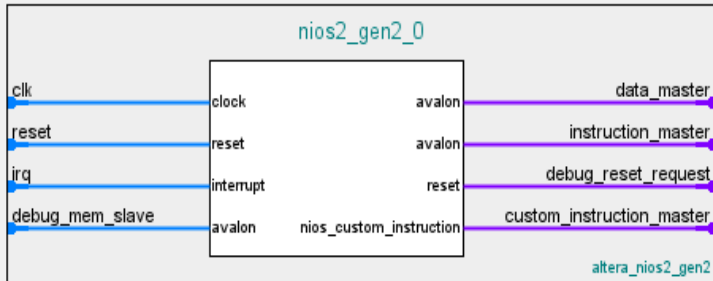
Nios II Processor - nios2\_gen2\_0



Nios II Processor  
altera\_nios2\_gen2

## Block Diagram

☐ Show signals



Main Vectors Caches and Memory Interfaces Arithmetic Instructions MMU and MPU Settings JTAG Debug Advanced F

## Select an Implementation

Nios II Core: ☐ Nios II/e  
☒ Nios II/f

|           | Nios II/e                        | Nios II/f                                                                                                                                                                                 |
|-----------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summary   | Resource-optimized 32-bit RISC   | Performance-optimized 32-bit RISC                                                                                                                                                         |
| Features  | JTAG Debug<br>ECC RAM Protection | JTAG Debug<br>Hardware Multiply/Divide<br>Instruction/Data Caches<br>Tightly-Coupled Masters<br>ECC RAM Protection<br>External Interrupt Controller<br>Shadow Register Sets<br>MPU<br>MMU |
| RAM Usage | 2 + Options                      | 2 + Options                                                                                                                                                                               |

# Creación System Hardware

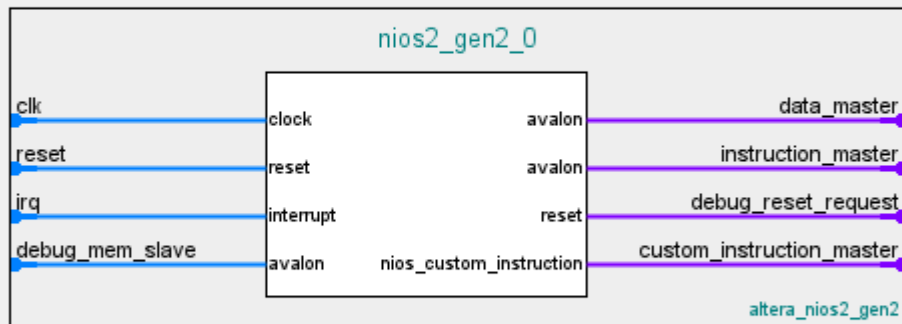
Nios II Processor - nios2\_gen2\_0



Nios II Processor  
altera\_nios2\_gen2

## Block Diagram

☐ Show signals



Main Vectors Caches and Memory Interfaces Arithmetic Instructions

## Select an Implementation

Nios II Core: ☒ Nios II/e  
☐ Nios II/f

|           | Nios II/e                        |
|-----------|----------------------------------|
| Summary   | Resource-optimized 32-bit RISC   |
| Features  | JTAG Debug<br>ECC RAM Protection |
| RAM Usage | 2 + Options                      |

# Creación System Hardware

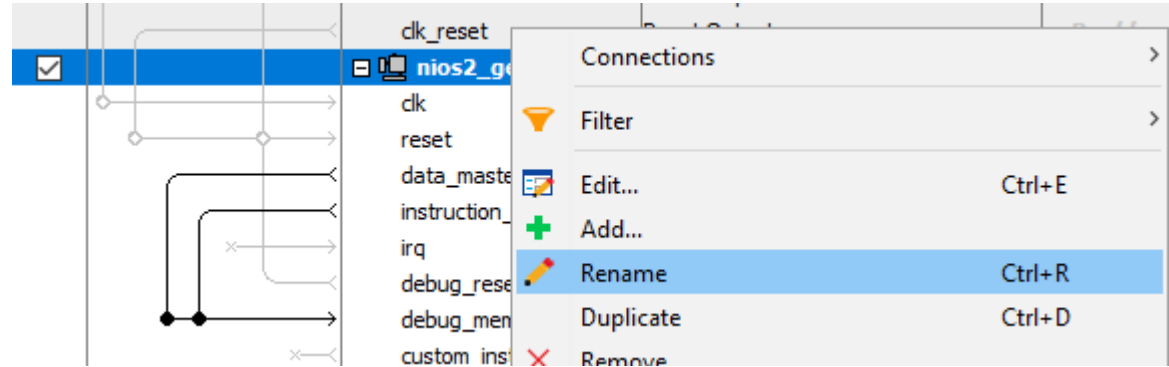
System Contents   Address Map   Interconnect Requirements

System: My\_firs\_Nios   Path: nios2\_gen2\_0

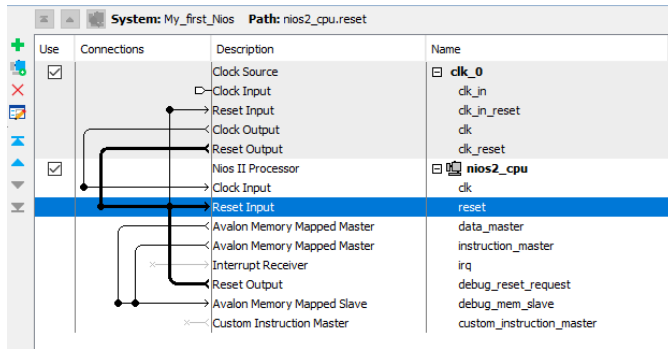
| Use                                 | Connections             | Name                       | Description                   | Export                        | Clock              | Base |
|-------------------------------------|-------------------------|----------------------------|-------------------------------|-------------------------------|--------------------|------|
| <input checked="" type="checkbox"/> |                         | clk_50                     | Clock Source                  |                               |                    |      |
|                                     |                         | clk_in                     | Clock Input                   | clk                           | <i>exported</i>    |      |
|                                     |                         | clk_in_reset               | Reset Input                   | reset                         |                    |      |
|                                     |                         | clk                        | Clock Output                  | <i>Double-click to export</i> | clk_50             |      |
|                                     |                         | clk_reset                  | Reset Output                  | <i>Double-click to export</i> |                    |      |
| <input checked="" type="checkbox"/> |                         | nios2_gen2_0               | Nios II Processor             |                               |                    |      |
|                                     |                         | clk                        | Clock Input                   | <i>Double-click to export</i> | <i>unconnected</i> |      |
|                                     |                         | reset                      | Reset Input                   | <i>Double-click to export</i> | [clk]              |      |
|                                     |                         | data_master                | Avalon Memory Mapped Master   | <i>Double-click to export</i> | [clk]              |      |
|                                     |                         | instruction_master         | Avalon Memory Mapped Master   | <i>Double-click to export</i> | [clk]              |      |
|                                     | irq                     | Interrupt Receiver         | <i>Double-click to export</i> | [clk]                         |                    |      |
|                                     | debug_reset_request     | Reset Output               | <i>Double-click to export</i> | [clk]                         |                    |      |
|                                     | debug_mem_slave         | Avalon Memory Mapped Slave | <i>Double-click to export</i> | [clk]                         |                    |      |
|                                     | custom_instruction_m... | Custom Instruction Master  | <i>Double-click to export</i> |                               |                    |      |

0x080

# Creación System Hardware



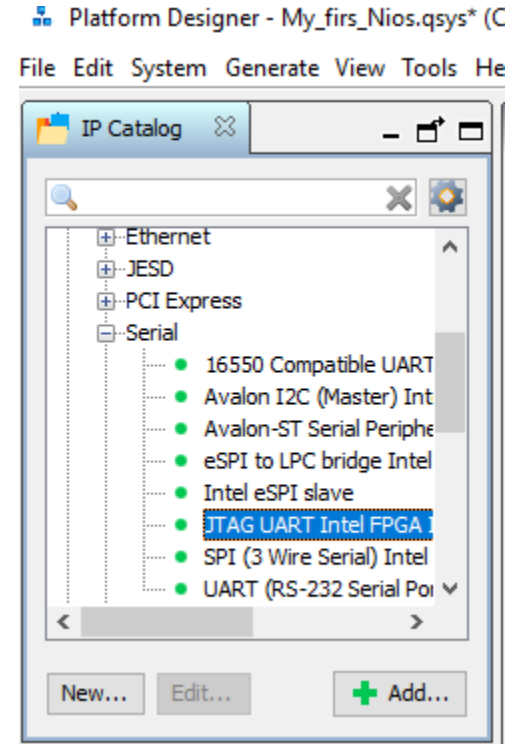
Renombrar los componentes, por ej; `nios2_gen2_0` -> `nios2`



Rutear el clock a los distintos componentes del sistema

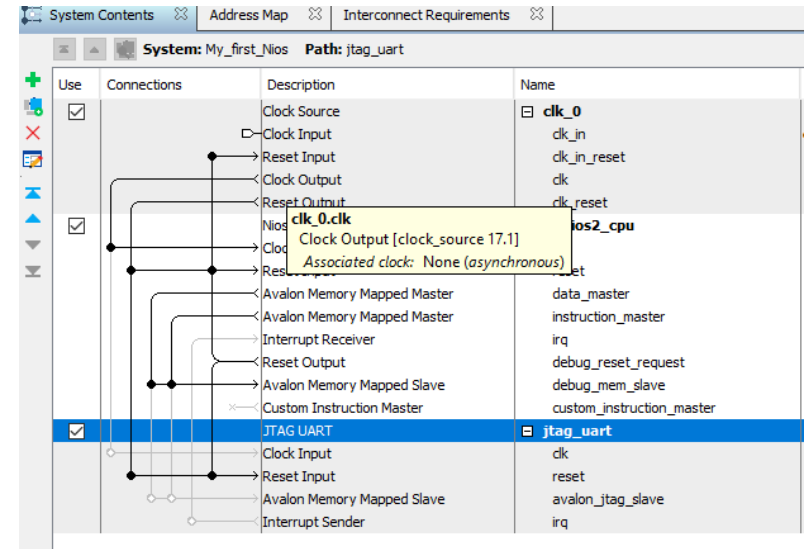
# Creación System Hardware

## ● Sumar un JTAG UART



# Creación System Hardware

- Unir las señales reset desde el CLOCK al módulo JTAG



# Creación System Hardware

- Sumar módulo de memoria (Size 20k)

**Size**

☐ Enable different width for Dual-port access

Slave S1 Data width: 32

Total memory size: 20480 bytes

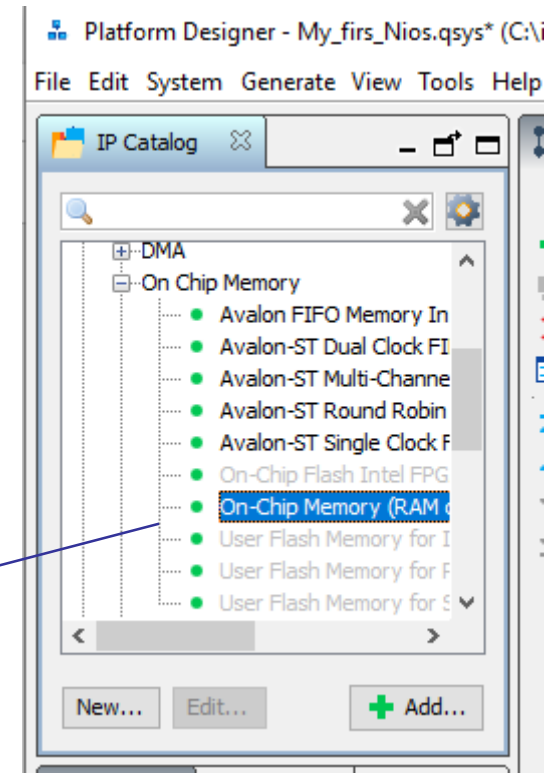
☐ Minimize memory block usage (may impact fmax)

**Read latency**

Slave s1 Latency: 1

Slave s2 Latency: 1

**ROM/RAM Memory Protection**





# Creación System Hardware

Asignar el “Reset Vector” y “Exception Vector” del Nios a la memoria”

The screenshot displays the Altera Nios II System Editor interface. The left pane, titled "System Cont", shows a hierarchical tree of system components. The right pane, titled "Parameters", shows the configuration for the "Nios II Processor".

**System Cont Window:**

| Use                                 | Connections | Name                    | Description     |
|-------------------------------------|-------------|-------------------------|-----------------|
| <input checked="" type="checkbox"/> |             | <b>clk_50</b>           | Clock Source    |
|                                     |             | clk_in                  | Clock Input     |
|                                     |             | clk_in_reset            | Reset Input     |
|                                     |             | clk                     | Clock Output    |
|                                     |             | clk_reset               | Reset Output    |
| <input checked="" type="checkbox"/> |             | <b>nios2_qsys</b>       | Nios II Process |
|                                     |             | clk                     | Clock Input     |
|                                     |             | reset                   | Reset Input     |
|                                     |             | data_master             | Avalon Memory   |
|                                     |             | instruction_master      | Avalon Memory   |
|                                     |             | irq                     | Interrupt Rece  |
|                                     |             | debug_reset_request     | Reset Output    |
|                                     |             | debug_mem_slave         | Avalon Memory   |
|                                     |             | custom_instruction_m... | Custom Instruc  |
| <input checked="" type="checkbox"/> |             | <b>jtag_uart_0</b>      | JTAG UART Int   |
|                                     |             | clk                     | Clock Input     |
|                                     |             | reset                   | Reset Input     |
|                                     |             | avalon_jtag_slave       | Avalon Memory   |
|                                     |             | irq                     | Interrupt Send  |
| <input checked="" type="checkbox"/> |             | <b>onchip_memory</b>    | On-Chip Memori  |

**Parameters Window:**

System: My\_firs\_Nios Path: nios2\_qsys

### Nios II Processor

altera\_nios2\_gen2

Main Vectors Caches and Memory Interfaces Arithmetic Instructions MMU and MPU Settings JTAG Debug Adv

**Reset Vector**

- Reset vector memory: None
- Reset vector offset: 0x00000000
- Reset vector: 0x00000000

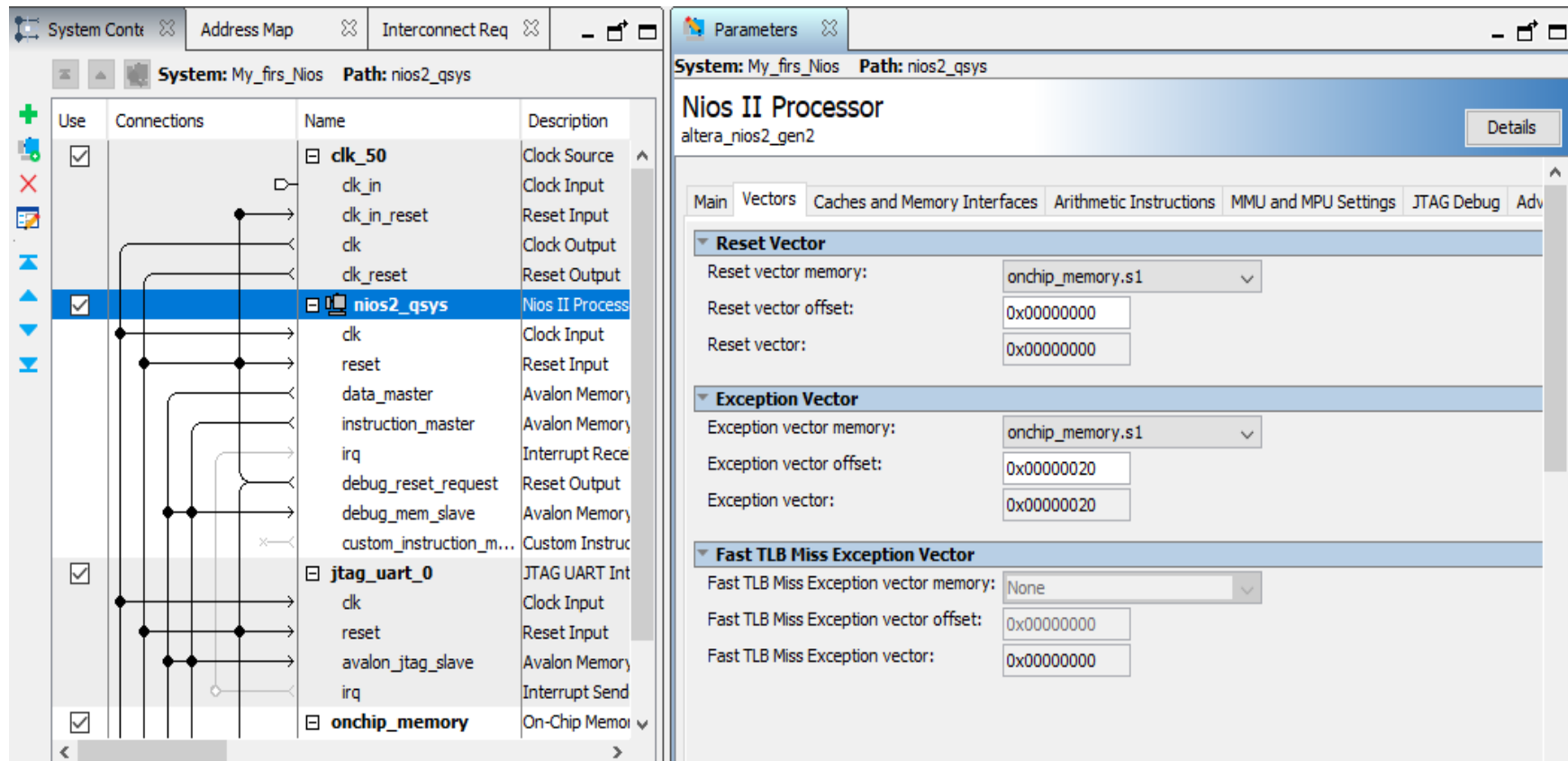
**Exception Vector**

- Exception vector memory: None
- Exception vector offset: 0x00000020
- Exception vector: 0x00000000

**Fast TLB Miss Exception Vector**

- Fast TLB Miss Exception vector memory: None
- Fast TLB Miss Exception vector offset: 0x00000000
- Fast TLB Miss Exception vector: 0x00000000

# Creación System Hardware



The screenshot displays the System Builder software interface, showing the hardware configuration for a Nios II processor. The left pane shows a list of components and their connections, while the right pane shows the parameters for the Nios II Processor.

**System: My\_firs\_Nios Path: nios2\_qsys**

**Use Connections**

| Use                                 | Connections | Name                    | Description     |
|-------------------------------------|-------------|-------------------------|-----------------|
| <input checked="" type="checkbox"/> |             | <b>clk_50</b>           | Clock Source    |
|                                     |             | clk_in                  | Clock Input     |
|                                     |             | clk_in_reset            | Reset Input     |
|                                     |             | clk                     | Clock Output    |
|                                     |             | clk_reset               | Reset Output    |
| <input checked="" type="checkbox"/> |             | <b>nios2_qsys</b>       | Nios II Process |
|                                     |             | clk                     | Clock Input     |
|                                     |             | reset                   | Reset Input     |
|                                     |             | data_master             | Avalon Memory   |
|                                     |             | instruction_master      | Avalon Memory   |
|                                     |             | irq                     | Interrupt Rece  |
|                                     |             | debug_reset_request     | Reset Output    |
|                                     |             | debug_mem_slave         | Avalon Memory   |
|                                     |             | custom_instruction_m... | Custom Instruc  |
| <input checked="" type="checkbox"/> |             | <b>jtag_uart_0</b>      | JTAG UART Int   |
|                                     |             | clk                     | Clock Input     |
|                                     |             | reset                   | Reset Input     |
|                                     |             | avalon_jtag_slave       | Avalon Memory   |
|                                     |             | irq                     | Interrupt Send  |
| <input checked="" type="checkbox"/> |             | <b>onchip_memory</b>    | On-Chip Memoi   |

**Parameters**

**System: My\_firs\_Nios Path: nios2\_qsys**

**Nios II Processor**  
altera\_nios2\_gen2

**Main Vectors Caches and Memory Interfaces Arithmetic Instructions MMU and MPU Settings JTAG Debug Adv**

**Reset Vector**

Reset vector memory: onchip\_memory.s1

Reset vector offset: 0x00000000

Reset vector: 0x00000000

**Exception Vector**

Exception vector memory: onchip\_memory.s1

Exception vector offset: 0x00000020

Exception vector: 0x00000020

**Fast TLB Miss Exception Vector**

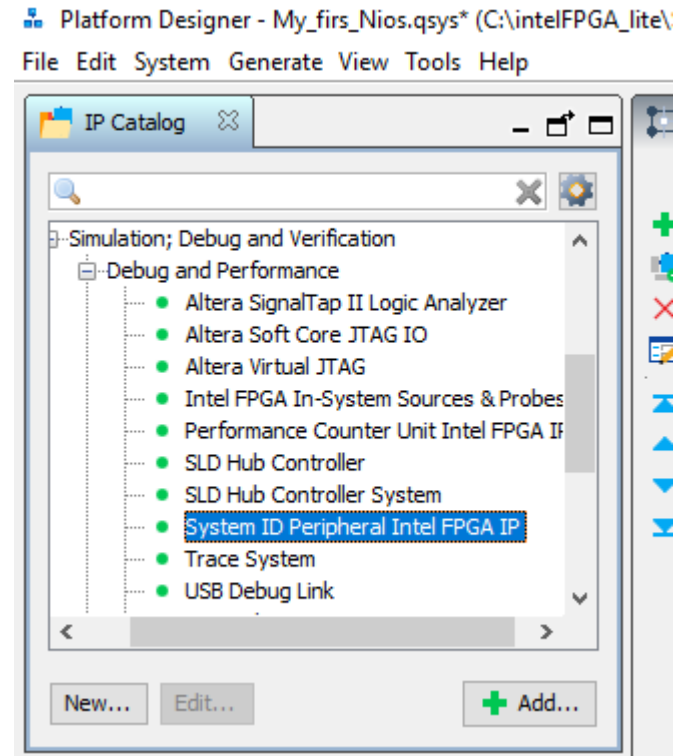
Fast TLB Miss Exception vector memory: None

Fast TLB Miss Exception vector offset: 0x00000000

Fast TLB Miss Exception vector: 0x00000000

# Creación System Hardware

- Sumar el periférico System ID



# Creación System Hardware

System ID Peripheral Intel FPGA IP - sysid\_qsys\_0

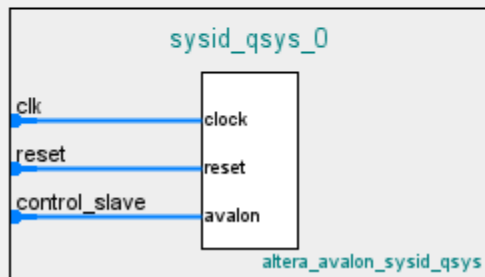


## System ID Peripheral Intel FPGA IP

altera\_avalon\_sysid\_qsys

### Block Diagram

☐ Show signals



### Parameters

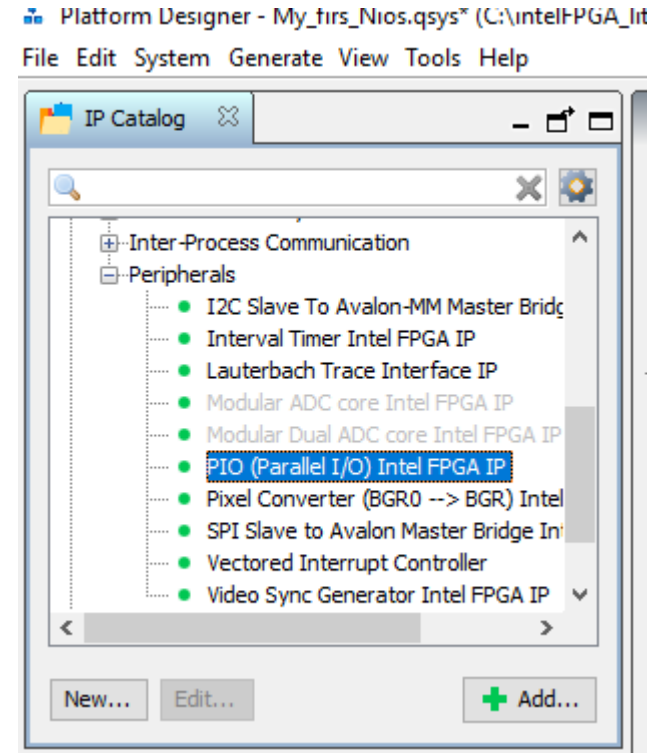
32 bit System ID:

### Description

Please use hexadecimal numbers only in System ID.

# Creación System Hardware

- Sumar el puerto de salida (renombrarlo como "leds")
- Sumar el puerto de entrada (renombrarlo como "switches")

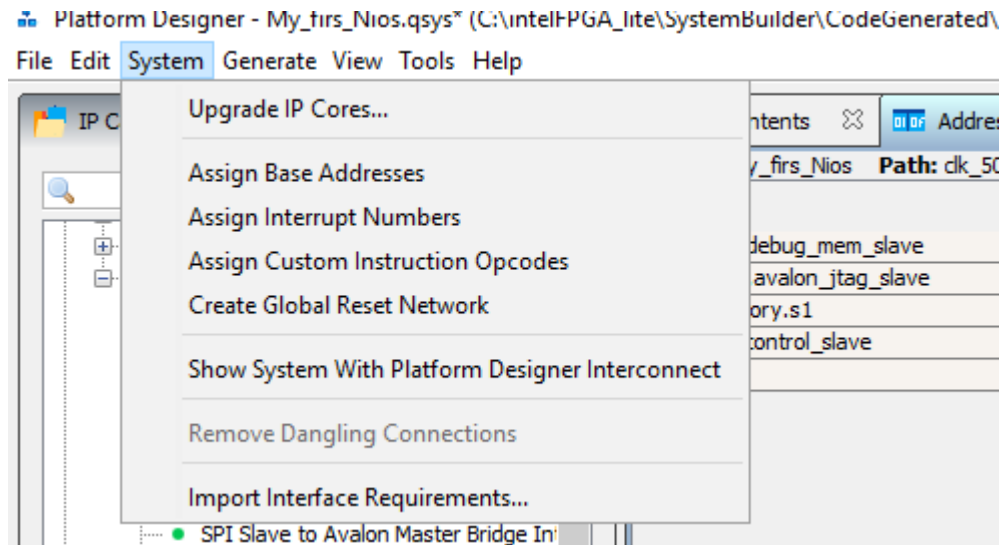


# Creación System Hardware

## Ajustar mapa de memoria

| Address Map                 |                       |  | Interconnect Requirements |                              |  |
|-----------------------------|-----------------------|--|---------------------------|------------------------------|--|
| System: My_first_Nios       |                       |  | Path: switches.dk         |                              |  |
|                             | nios2_cpu.data_master |  |                           | nios2_cpu.instruction_master |  |
| nios2_cpu.debug_mem_slave   | ✗ 0x0800 - 0x0fff     |  | ✗ 0x0800 - 0x0fff         |                              |  |
| jtag_uart.avalon_jtag_slave | ✗ 0x0000 - 0x0007     |  | ✗ 0x0000 - 0x0007         |                              |  |
| onchip_memory.s1            | ✗ 0x0000 - 0x4fff     |  | ✗ 0x0000 - 0x4fff         |                              |  |
| sysid_qsys.control_slave    | ✗ 0x0000 - 0x0007     |  | ✗ 0x0000 - 0x0007         |                              |  |
| leds.s1                     | ✗ 0x0000 - 0x000f     |  | ✗ 0x0000 - 0x000f         |                              |  |
| switches.s1                 | ✗ 0x0000 - 0x000f     |  | ✗ 0x0000 - 0x000f         |                              |  |

# Creación System Hardware



# Creación System Hardware

| Type            | Path                       | Message                                                                                                      |
|-----------------|----------------------------|--------------------------------------------------------------------------------------------------------------|
| 2 Warnings      |                            |                                                                                                              |
|                 | My_first_Nios.leds         | leds.external_connection must be exported, or connected to a matching conduit.                               |
|                 | My_first_Nios.switches     | switches.external_connection must be exported, or connected to a matching conduit.                           |
| 4 Info Messages |                            | switches.external_connection must be exported, or connected to a matching conduit.                           |
|                 | My_first_Nios.jtag_uart    | JTAG UART IP input clock need to be at least double (2x) the operating frequency of JTAG TCK on board        |
|                 | My_first_Nios.sysid_qsys_0 | System ID is not assigned automatically. Edit the System ID parameter to provide a unique ID                 |
|                 | My_first_Nios.sysid_qsys_0 | Time stamp will be automatically updated when this component is generated.                                   |
|                 | My_first_Nios.switches     | PIO inputs are not hardwired in test bench. Undefined values will be read from PIO inputs during simulation. |



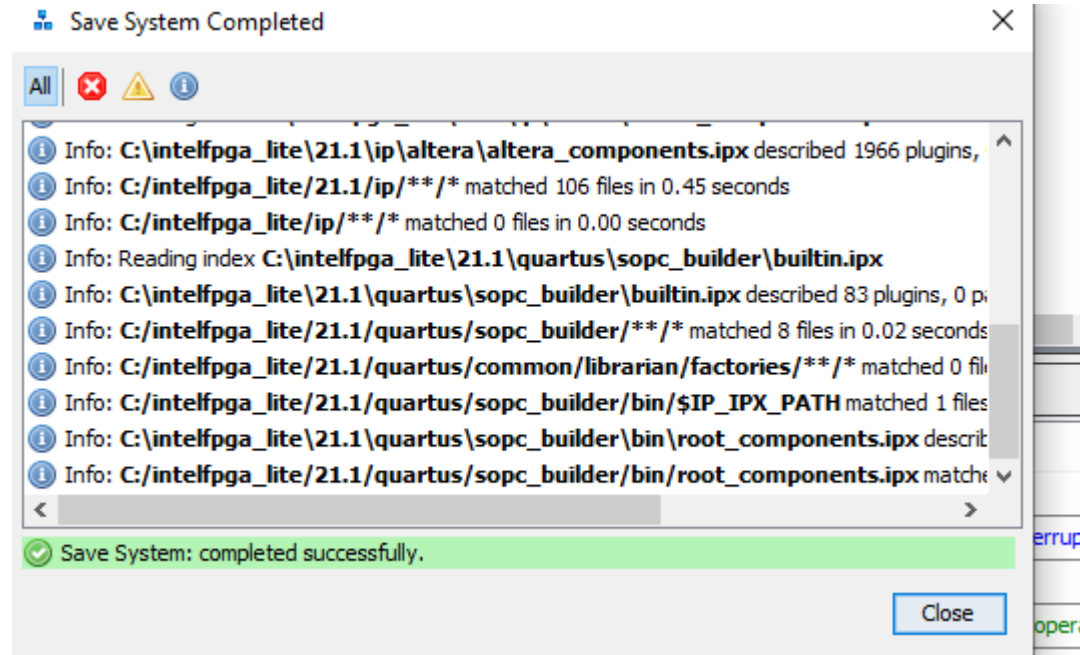
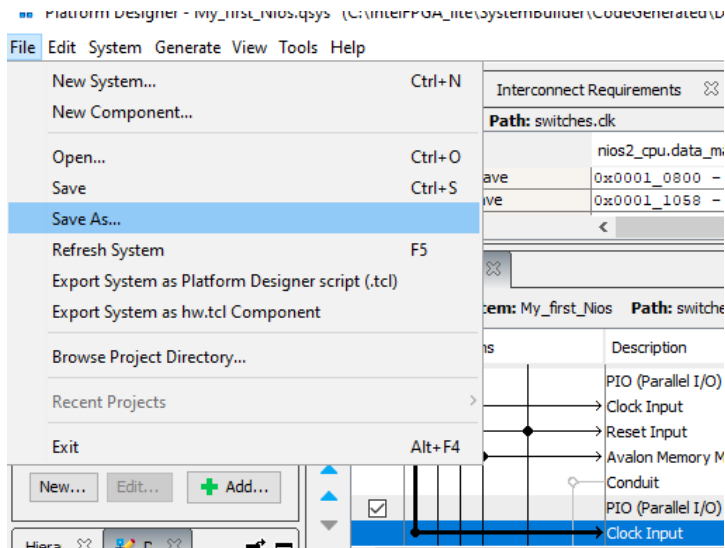
# Creación System Hardware

## Exportar leds y switches

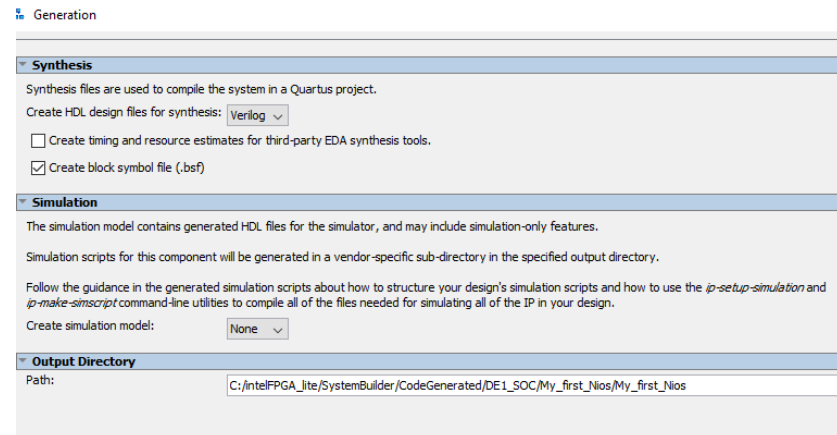
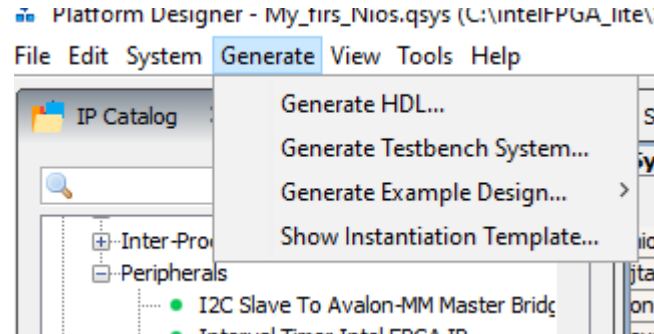
| Name                | Export                       | Clock | Base        | End    |
|---------------------|------------------------------|-------|-------------|--------|
| <b>leds</b>         |                              |       |             |        |
| clk                 | Double-click to export       | clk_0 |             |        |
| reset               | Double-click to export       | [clk] |             |        |
| s1                  | Double-click to export       | [clk] | 0x0001_1030 | 0x0001 |
| external_connection | leds_external_connection     |       |             |        |
| <b>switches</b>     |                              |       |             |        |
| clk                 | Double-click to export       | clk_0 |             |        |
| reset               | Double-click to export       | [clk] |             |        |
| s1                  | Double-click to export       | [clk] | 0x0001_1020 | 0x0001 |
| external_connection | switches_external_connection |       |             |        |

# Creación System Hardware

## Salvar el .qsys y generar el .sopcinfo



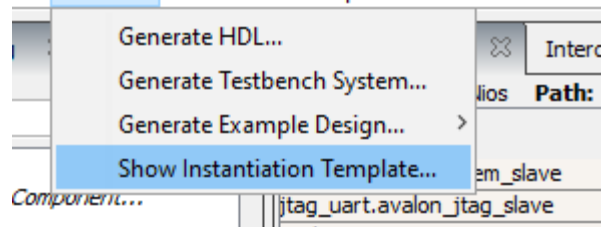
# Creación System Hardware



# Instanciar el SoPC en Quartus

designer - My\_first\_Nios.qsys (C:\intel\FPGA\_lite\SystemB

em Generate View Tools Help



## Instantiation Template

You can copy the example HDL below to declare an instance of **My\_first\_Nios**.

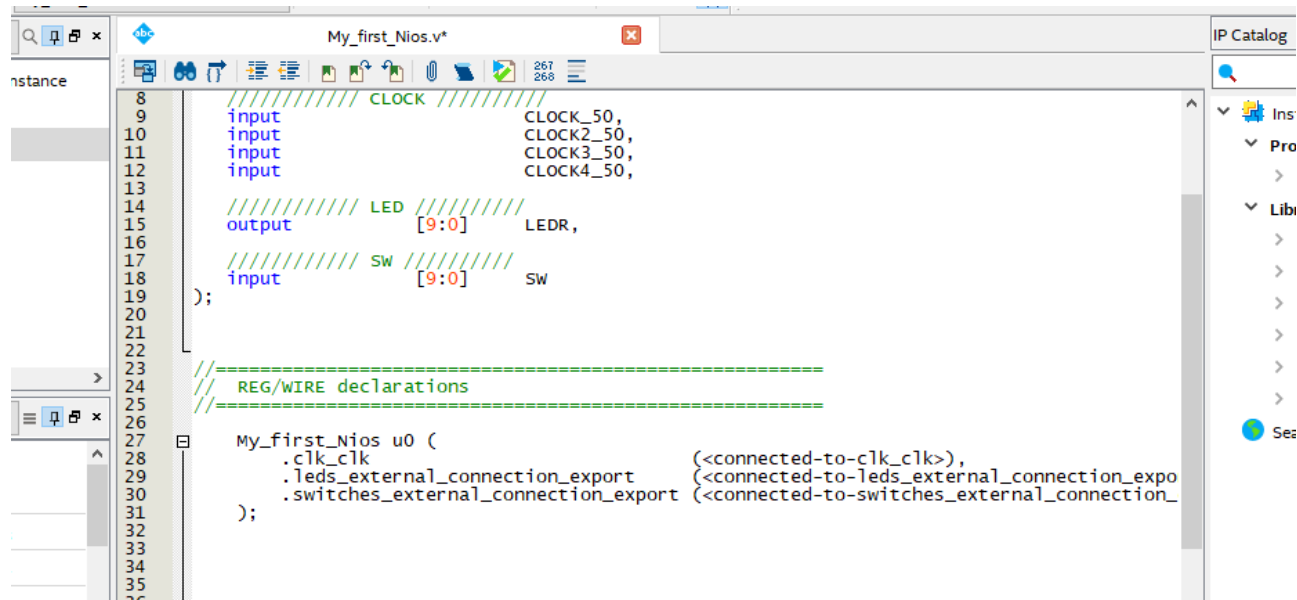
HDL Language: Verilog ▾

Example HDL

```
My_first_Nios u0 (  
    .clk_clk (<connected-to-clk_clk>), //  
    .leds_external_connection_export (<connected-to-leds_external_connection_export>), // leds_external_  
    .switches_external_connection_export (<connected-to-switches_external_connection_export>) // switches_external_  
);
```

# Instanciar el hardware

## Copiar en quartus el template



The screenshot displays the Quartus II IDE with a Verilog HDL file named 'My\_first\_Nios.v\*' open. The file contains the following code:

```
8 /////////////////////////////////////////////////// CLOCK ////////////////////////////////////////////
9 input                                CLOCK_50,
10 input                                CLOCK2_50,
11 input                                CLOCK3_50,
12 input                                CLOCK4_50,
13
14 /////////////////////////////////////////////////// LED ////////////////////////////////////////////
15 output [9:0] LEDR,
16
17 /////////////////////////////////////////////////// SW ////////////////////////////////////////////
18 input [9:0] SW
19
20 );
21
22
23 //=====
24 // REG/WIRE declarations
25 //=====
26
27 My_first_Nios u0 (
28     .clk_clk (<connected-to-clk_clk>),
29     .leds_external_connection_export (<connected-to-leds_external_connection_expo
30     .switches_external_connection_export (<connected-to-switches_external_connection_
31 );
32
33
34
35
36
```

The interface includes a search bar, a toolbar with various icons, and a right-hand pane titled 'IP Catalog' with a search icon and a list of categories: 'Ins', 'Pro', and 'Libi'. The 'Ins' category is expanded, showing a list of components with a search icon and a 'See' button.



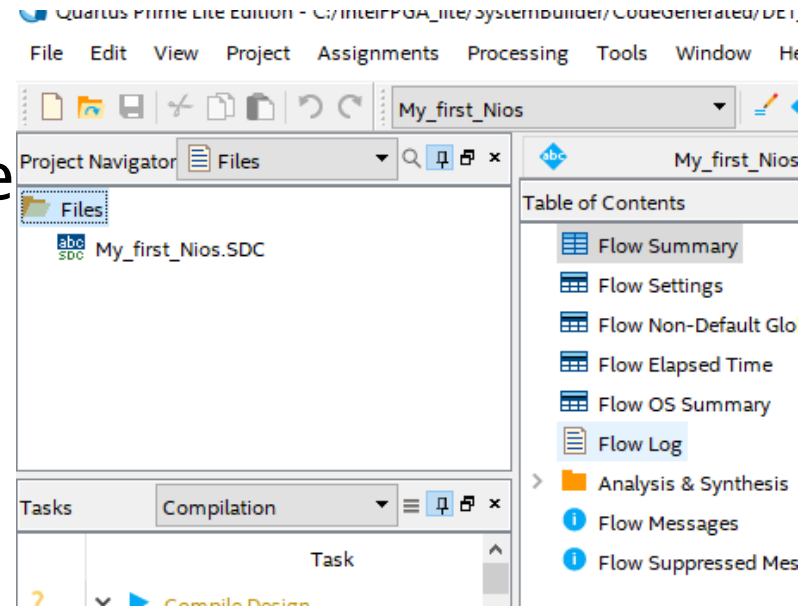
# Instanciar el hardware

## Asignar las entradas y salidas al nios

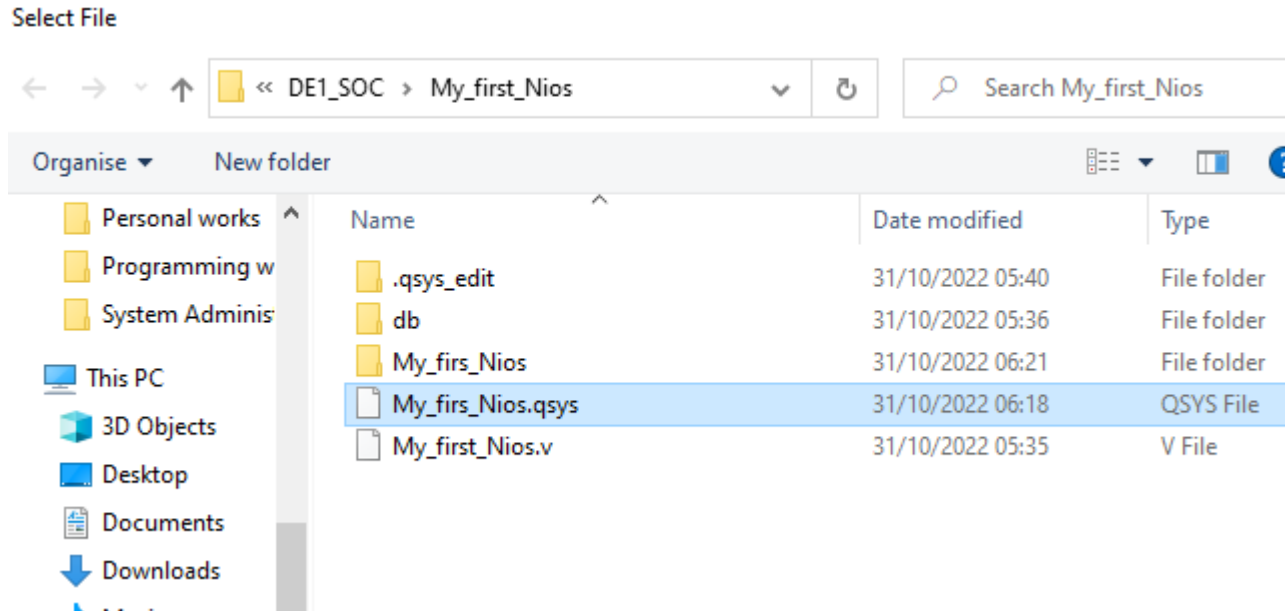
```
18 input [9:0] SW
19 );
20
21
22
23 //=====
24 // REG/WIRE declarations
25 //=====
26
27 My_first_Nios u0 (
28     .clk_clk (CLOCK_50),
29     .leds_external_connection_export (LEDR[7:0]), // leds_external.
30     .switches_external_connection_export (SW[7:0]) // switches_external_conne
31 );
32
33
34
35
36
37
38 //=====
39 // structural coding
40 //=====
```

# Instanciar el hardware

Antes de compilar el proyecto, ir a la parte de Files y sumar el .qsys generado por Platform Designer



# Instanciar el hardware



Después de sumarlo al proyecto ,  
compilar





# Instanciar el hardware

---

```
i 332123 Deriving Clock Uncertainty. Please refer to report_sdc in the Timing Analyzer
> i 332146 worst-case setup slack is 12.535
> i 332146 worst-case hold slack is 0.009
> i 332146 worst-case recovery slack is 17.736
> i 332146 worst-case removal slack is 0.336
> i 332146 worst-case minimum pulse width slack is 8.438
> i 332114 Report Metastability: Found 9 synchronizer chains.
i 332102 Design is not fully constrained for setup requirements
i 332102 Design is not fully constrained for hold requirements
> i          Quartus Prime Timing Analyzer was successful. 0 errors, 2 warnings
i 293000 Quartus Prime Full Compilation was successful. 0 errors, 28 warnings
```

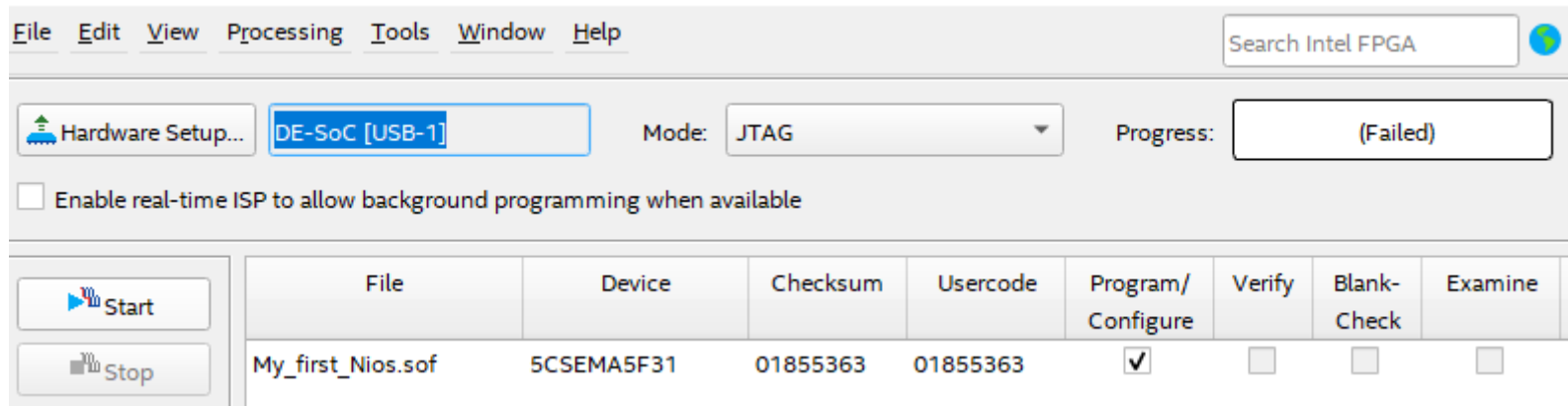
# Download hardware a FPGA

| File              | Device     | Checksum | Usercode | Program/Configure                   | Verify                   | Blank-Check              | Extra |
|-------------------|------------|----------|----------|-------------------------------------|--------------------------|--------------------------|-------|
| <none>            | SOCVHPS    | 00000000 | <none>   | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | [     |
| My_first_Nios.sof | 5CSEMA5F31 | 01B06C3C | 01B06C3C | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | [     |

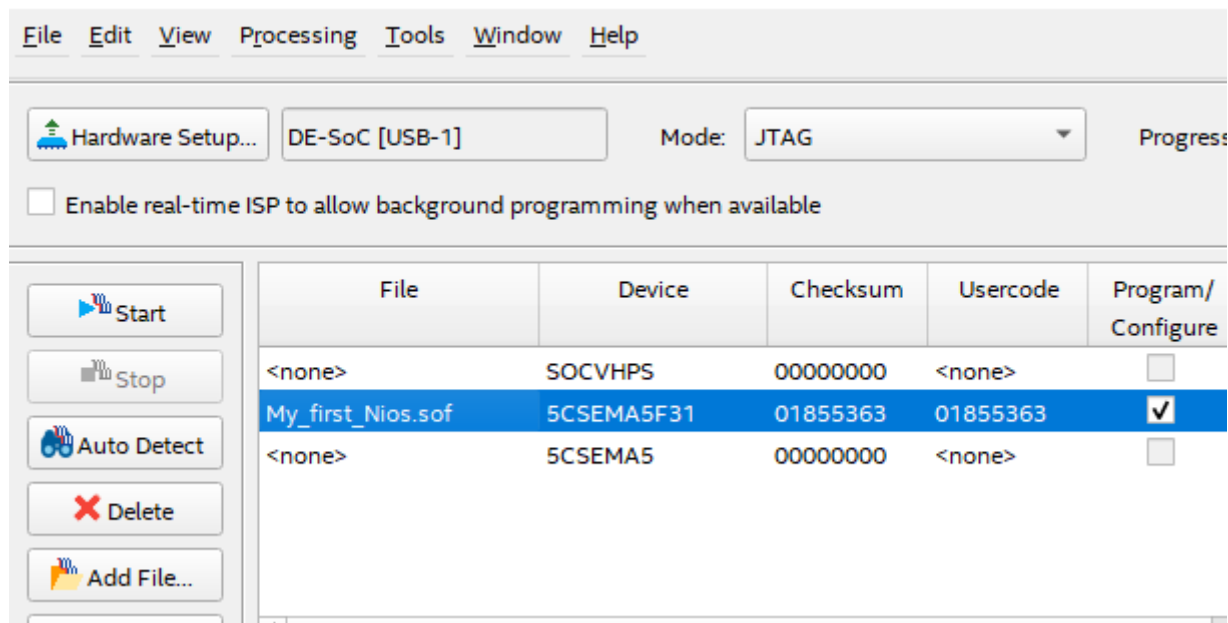
Diagram illustrating the hardware download process:

```
graph LR; TDI --> SOCVHPS; SOCVHPS --> 5CSEMA5F31; 5CSEMA5F31 --> TDO;
```

# Download hardware a FPGA



# Download hardware a FPGA



# Download hardware a FPGA

Programmer - C:/intelFPGA\_lite/SystemBuilder/CodeGenerated/DE1\_SOC/My\_first\_Nios/My\_first\_Nios - My\_first\_Nios...

File Edit View Processing Tools Window Help

Search Intel FPGA

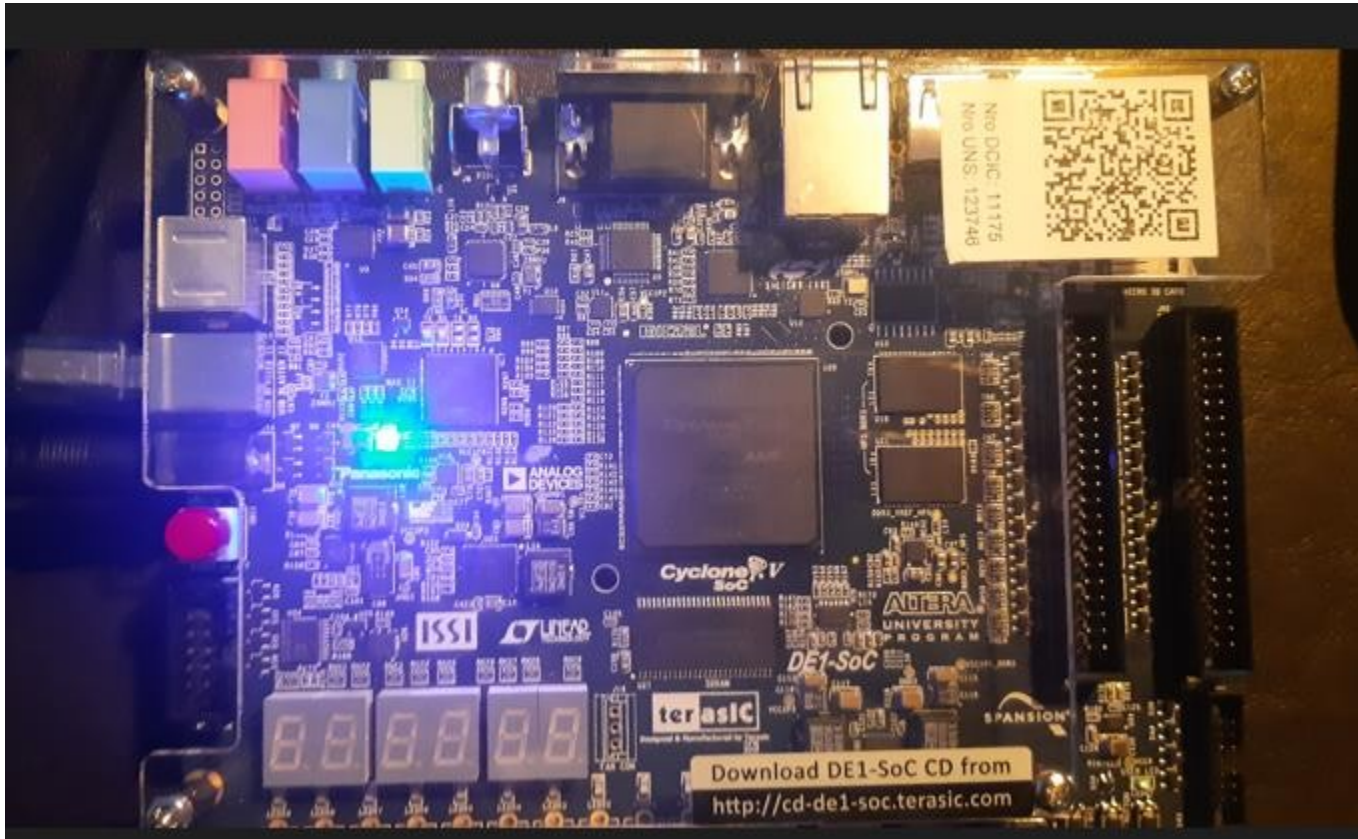
Hardware Setup... DE-SoC [USB-1] Mode: JTAG Progress: 100% (Successful)

☐ Enable real-time ISP to allow background programming when available

Start Stop Auto Detect Delete Add File...

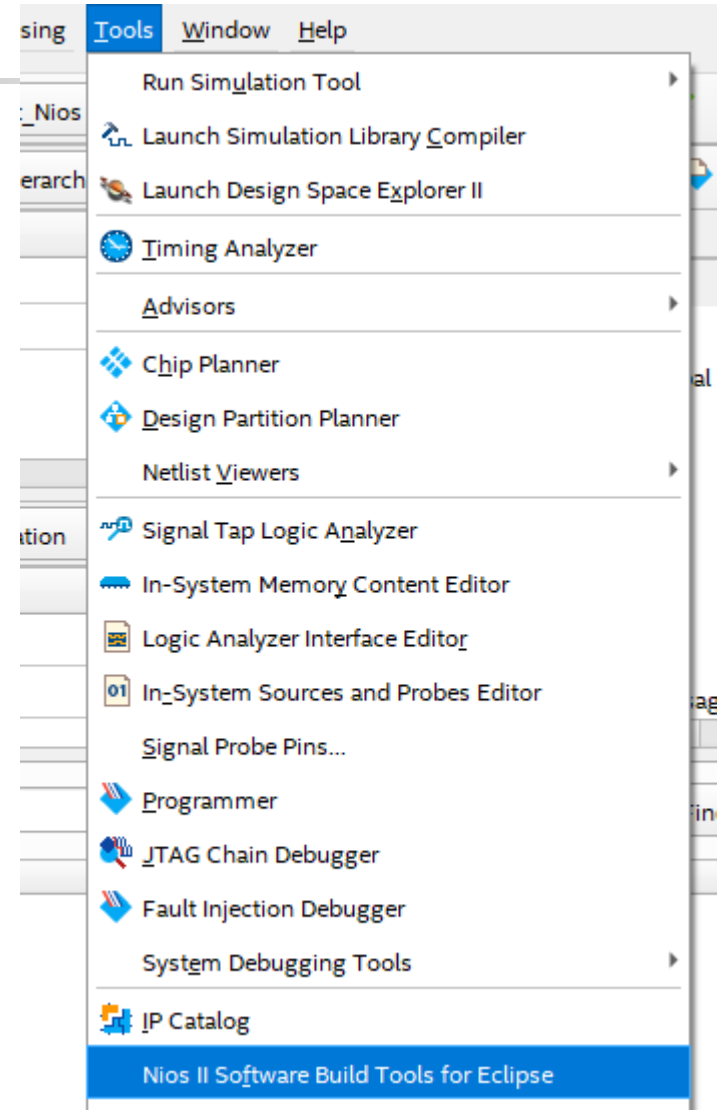
| File              | Device     | Checksum | Usercode | Program/Configure                   | Verify                   | Blank-Check              | Examine                  |
|-------------------|------------|----------|----------|-------------------------------------|--------------------------|--------------------------|--------------------------|
| <none>            | SOCVHPS    | 00000000 | <none>   | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| My_first_Nios.sof | 5CSEMA5F31 | 01855363 | 01855363 | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

# Download hardware a FPGA

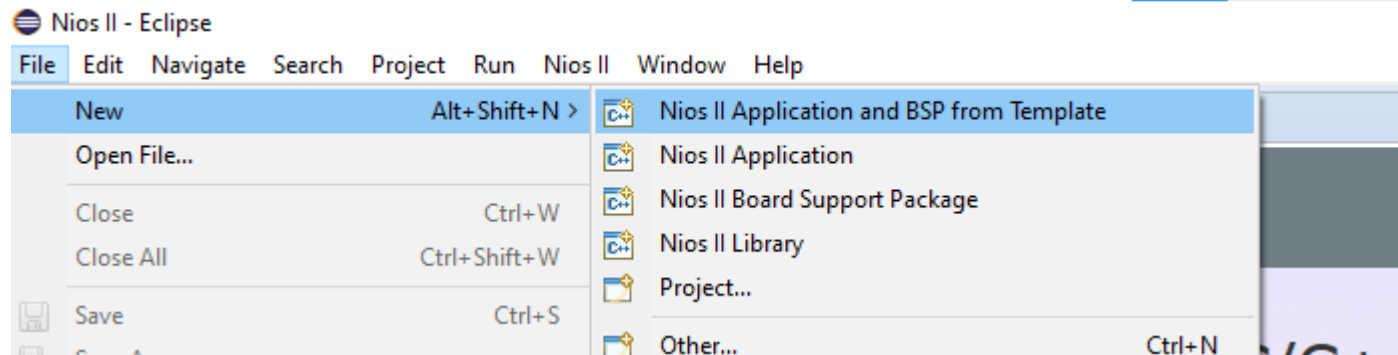


# Nios II Software Eclipse

- Crear proyecto para NiosII



# Nios II Software Eclipse



Seleccionar el .sopcinfo generado por el Platform Designer



# Nios II Software Eclipse

Target hardware information

SOPC Information File name:  ...

CPU name:

Application project

Project name:

☒ Use default location

Project location:  ...

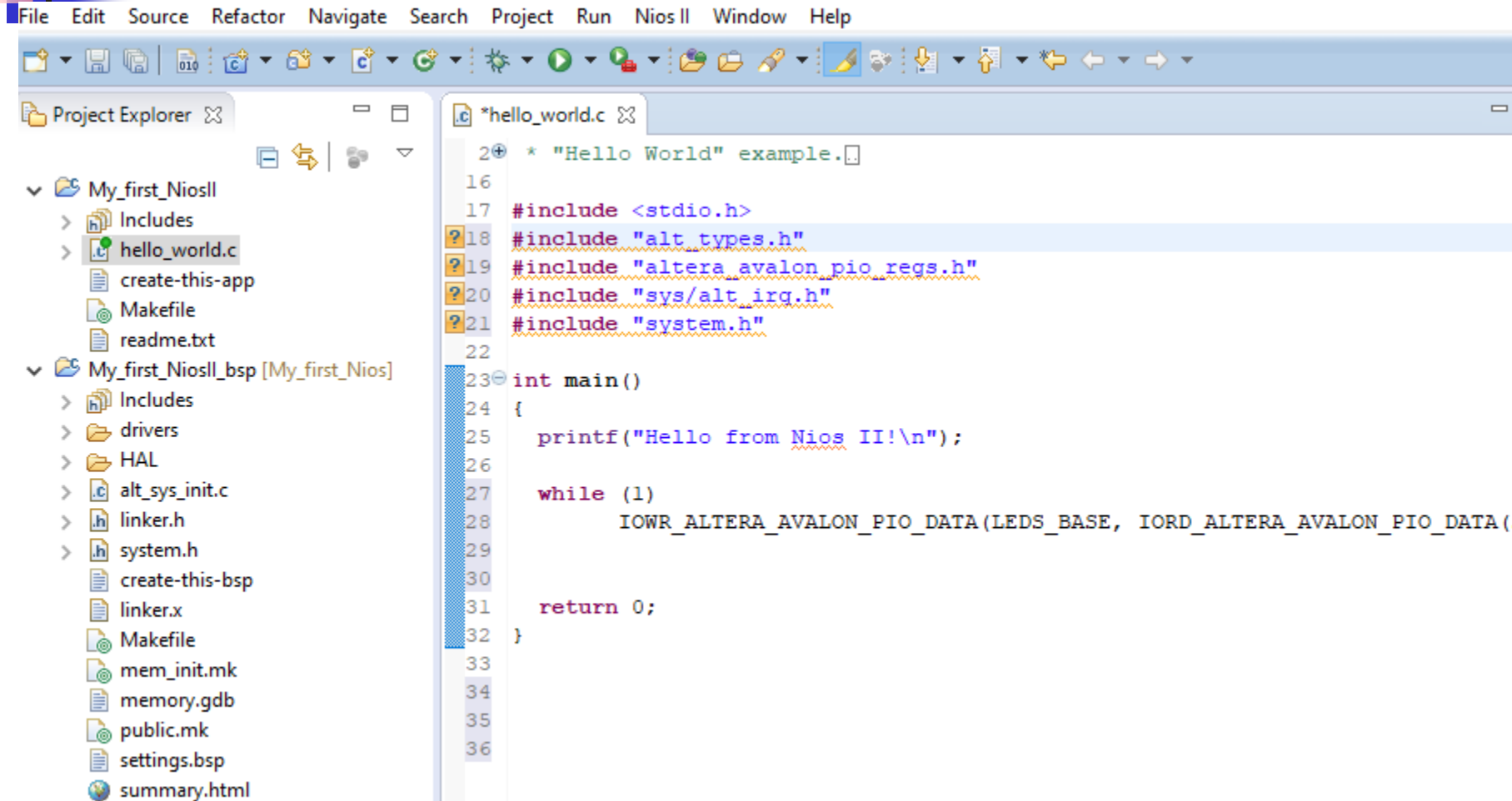
Project template

| Templates                                                                | Template description                                                     |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Blank Project                                                            | Blank Project                                                            |
| Blank Project with CPU                                                   | Blank Project with CPU                                                   |
| Blank Project with CPU and MMIO                                          | Blank Project with CPU and MMIO                                          |
| Blank Project with CPU and MMIO and GPIO                                 | Blank Project with CPU and MMIO and GPIO                                 |
| Blank Project with CPU and MMIO and GPIO and UART                        | Blank Project with CPU and MMIO and GPIO and UART                        |
| Blank Project with CPU and MMIO and GPIO and UART and Ethernet           | Blank Project with CPU and MMIO and GPIO and UART and Ethernet           |
| Blank Project with CPU and MMIO and GPIO and UART and Ethernet and Video | Blank Project with CPU and MMIO and GPIO and UART and Ethernet and Video |

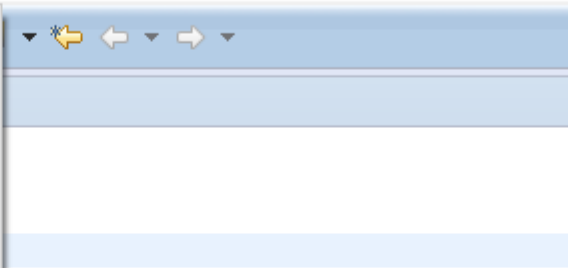
Pictures  
Videos

| File Name              | Size       | Created          | Modified         |
|------------------------|------------|------------------|------------------|
| My_first_Nios.sopcinfo | 1024 bytes | 06/11/2022 11:07 | 06/11/2022 11:07 |

# Nios II Software Eclipse



System timestamp mismatch - connected: "1509136575", expected: "1509385808".



## Construir el.elf y volcarlo a la FPGA



# Nios II Software Eclipse

---

- Si obtiene errores de timestamp, setee las opciones de mismatched

Create, manage, and run configurations

✖ System timestamp mismatch - connected: "1509136575", expected: "1509385808".

Quartus Project File name: < Using default .sopcinfo & .jdi files extracted from ELF >

## System ID checks

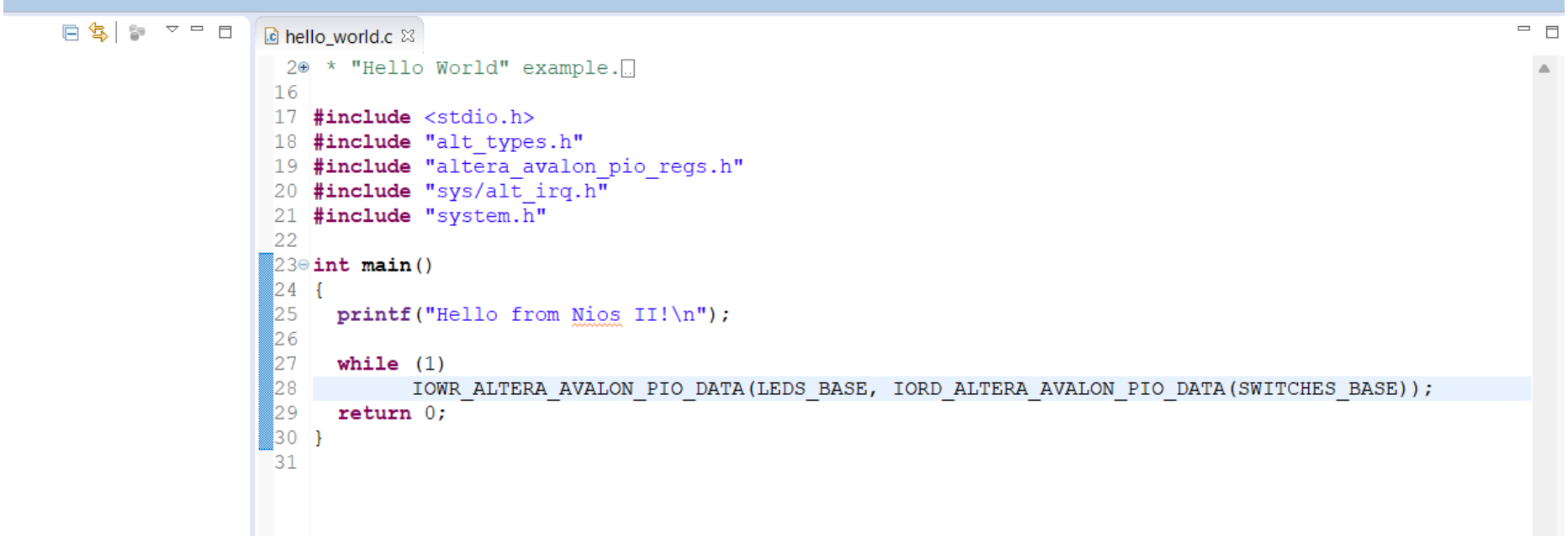
- ☒ Ignore mismatched system ID
- ☒ Ignore mismatched system timestamp

Download



# Nios II Software Eclipse

Determinar de dónde el sistema obtiene los valores de LEDS\_BASE y SWITCHES\_BASE en el main.c



```
20 * "Hello World" example.
16
17 #include <stdio.h>
18 #include "alt_types.h"
19 #include "altera_avalon_pio_regs.h"
20 #include "sys/alt_irq.h"
21 #include "system.h"
22
23 int main()
24 {
25     printf("Hello from Nios II!\n");
26
27     while (1)
28         IOWR_ALTERA_AVALON_PIO_DATA(LED_BASE, IORD_ALTERA_AVALON_PIO_DATA(SWITCHES_BASE));
29     return 0;
30 }
31
```

Corroborar el cambio de la entrada (SW) con respecto a la salida (LEDS) en la fpga.

# Nios II Software Eclipse

